Achieving Envy-Freeness with Limited Subsidies under Negative Dichotomous Valuations

Mainak Sarkar

Abstract

We study the problem of allocating indivisible *chores* among agents in a fair manner, with subsidies. Envy-free allocations of indivisible items are not guaranteed to exist, but envy-freeness can be achieved by additionally providing some subsidy to the agents; in the chores setting this subsidy is naturally read as compensation paid to the agents who absorb the workload. We study the subsidy required under *negative dichotomous* valuations, i.e., among agents whose marginal disutility for any chore is either zero or one: $v_i(S) - v_i(S \cup \{g\}) \in \{0, 1\}$ for every agent i, every $S \subseteq M$ and every $g \in M \setminus S$. This is the exact mirror of the dichotomous goods model of Barman et al. [BKNS22], who prove that there a per-agent subsidy of 0 or 1 always suffices, and we conjecture that the same guarantee holds for chores.

Existing work already gives a bounded subsidy: negative dichotomous valuations are doubly monotone under the two-sided normalisation, so the reduction of Kawase et al. [KMS+24] yields $n - 1$ per agent and $n(n - 1)/2$ in total. The question is whether the far stronger $\{0, 1\}$ guarantee of [BKNS22] survives the passage from goods to chores, a factor of n below that baseline. The Halpern–Shah characterisation of envy-freeable allocations [HS19] transfers verbatim, since it assumes nothing about the sign or the structure of the valuations, and it delivers an *integral* minimum subsidy vector here; the matching lower bound of $n - 1$ transfers as well, so the conjectured bound would be tight.

We prove the conjecture for three agents: every instance with $n = 3$ admits an envy-free solution with a subsidy of 0 or 1 per agent, hence at most 2 in total, computable in polynomial time in the value-oracle model, and under no assumption on the cost functions beyond binary marginals — not additivity, not submodularity, and no bound on the number of chores. The proof completes an envy-free *partial* allocation, taken from Tao et al. [TWYZ23], by placing the residual chores and reassigning which agent receives which bundle; the subsidy bound follows from a bound on the total excess of that reassignment.

1 Introduction

Discrete fair division studies the allocation of resources that cannot be fractionally assigned among agents with individual preferences, and sits at the interface of mathematical economics and computer science [BCE+16, End17]. The classical fairness criterion is *envy-freeness* [Fol67, Var74]: no agent should prefer the bundle assigned to another agent over her own. For indivisible items an envy-free allocation need not exist, as the standard one-item two-agent instance shows, and a large body of work therefore studies relaxations such as envy-freeness up to one good (EF1) [Bud11, LMMS04].

A second and complementary line of work retains *exact* envy-freeness and buys off the residual envy with money. Each agent i receives, in addition to her bundle, a subsidy $p_i \geq 0$ supplied by a third party, and one asks for an allocation together with a subsidy vector under which every agent weakly prefers her own bundle-plus-subsidy to that of every other agent. The question is how much money this requires. Maskin [Mas87] answered it for unit-demand agents with $m = n$: under the scaling in which no agent values any single item above 1, a total subsidy of $n-1$ suffices. Halpern and Shah [HS19] moved to the multi-demand setting with arbitrary m, gave the characterisation of *envy-freeable* allocations that the whole line now runs on, showed that a subsidy of $(n-1)m$ is necessary and sufficient in the worst case when the allocation is handed to us, and conjectured that $n-1$ suffices when we may choose the allocation. Brustle et al. [BDN+20] proved that conjecture in the stronger per-agent form — one dollar to each agent suffices for additive valuations, with the allocation simultaneously EF1 and balanced — and showed that for general monotone valuations $2(n-1)$ per agent suffices, so that the total subsidy is $O(n^2)$ and, remarkably, *independent of the number of items*. Barman et al. [BKNS22] then improved the monotone bound by a factor of n on the dichotomous subclass: when every marginal $v_i(S \cup \{g\}) - v_i(S)$ lies in $\{0, 1\}$, there is an allocation whose accompanying subsidy is exactly 0 or 1 for each agent, hence at most $n-1$ in total, computable in polynomial time in the value-oracle model. Their bound assumes neither additivity nor submodularity nor even subadditivity, and it is tight: with n agents and a single unit-valued good, $n-1$ agents must each be paid 1.

Chores. Every result above is about *goods*. This paper asks the mirrored question, in which every item is a *chore*: an object that an agent would rather not hold, so that acquiring it weakly decreases her utility. Chores are not an exotic variant. Shift rotations, on-call duty, teaching and refereeing load, committee service, maintenance tasks and the liabilities attached to an estate are all allocation problems in which the items are burdens rather than prizes, and in all of them money is already the standard instrument of repair — as overtime pay, hardship allowance, or a course-release budget. If anything the subsidy model is more natural for chores than for goods, since compensating somebody for taking on unwanted work needs less justification than paying somebody for the privilege of not receiving a prize.

Concretely, we study *negative dichotomous* valuations: for every agent i, every $S \subseteq M$ and every $g \in M \setminus S$,

$$v_i(S) - v_i(S \cup \{g\}) \in \{0, 1\}.$$

Equivalently, each marginal $v_i(S \cup \{g\}) - v_i(S)$ is 0 or -1: every item is a chore for every agent, and taking on one more chore costs an agent either nothing or exactly one unit. This is the exact mirror of the model of [BKNS22], and we impose no further structure — in particular the valuations need not be additive, submodular, or subadditive.

Binary marginals are the natural model whenever a task either lands on an agent as a fresh unit of burden or is absorbed by a commitment she has already made. An engineer already staffing an on-call window absorbs a second incident in that window at no extra cost; a referee already reading for a venue absorbs one more paper from the same batch; a driver already routed through a district absorbs an extra stop along the way. Whether a given chore is free depends in an arbitrary way on the rest of the agent's assignment, which is exactly why the model must tolerate non-additive and even non-subadditive valuations, and it is the counterpart of the scheduling example used by [BKNS22] to motivate dichotomous valuations for goods.

What is already known, and what is open. Two facts delimit the problem before any work is done, and we state them explicitly so that the contribution can be measured against them.

On the one hand, existence of a bounded subsidy is not in question. Negative dichotomous valuations are doubly monotone — every item is a chore for every agent, which is a special case of every item being a good or a chore for each agent — and they satisfy the normalisation $|v_i(S \cup \{e\}) - v_i(S)| \leq 1$ that [KMS+24, §2] imposes. The reduction of Kawase et al., which converts any EF1 allocation into an envy-free one, therefore applies off the shelf and yields a subsidy of at most $n-1$ per agent and $n(n-1)/2$ in total [KMS+24, Theorem 1]. Two details make the containment work and are easy to get wrong. Their normalisation is two-sided, so it bounds chore disutility as well as value; and the EF1 they require is the two-sided form, which removes one item from *either* bundle, not the goods-only form of [HS19] — the two differ precisely when chores are present. An EF1 allocation is guaranteed to exist and to be computable in polynomial time for doubly monotone valuations, so the reduction has an input; for that step [KMS+24, §2] cites [LMMS04] and [BSV21], and the citation matters, because the envy-cycle argument of [LMMS04] does *not* port to chores unaltered and it is [BSV21] that supplies the repair.

On the other hand, the lower bound of the goods model survives intact.

Example 1 (The $n-1$ lower bound transfers)**.** Take n agents and $m = n-1$ chores, with $c_i(S) = |S|$ for every agent i — identical, binary, additive costs, which are in particular negative dichotomous. Every complete allocation leaves at least one agent empty-handed. Under the balanced allocation, in which $n-1$ agents hold one chore each and one agent holds nothing, each of the $n-1$ loaded agents envies the empty agent by exactly 1 and no other envy is present, so the minimum subsidy assigns 1 to each loaded agent and 0 to the empty one, for a total of $n-1$. No allocation does better: any complete allocation gives some agent a bundle of size $k \geq 1$ while some other agent is empty, and that agent's heaviest outgoing path already has weight k.

So the target statement is precisely the analogue of [BKNS22, Theorem 4].

Conjecture 2. *For every instance with negative dichotomous valuations there is an envy-free solution $(\mathcal{A}, p)$ with $p \in \{0,1\}^n$, hence with total subsidy at most $n-1$; and it can be computed in polynomial time given value-oracle access.*

Example 1 shows Conjecture 2 would be tight. Between the $n-1$ per agent that [KMS+24] already gives and the $\{0,1\}$ per agent being asked for lies a factor of n in the total, exactly the gap that [BKNS22] closed on the goods side. Our objective is to close it on the chores side or to exhibit an instance showing that it cannot be closed.

Our contribution. We prove Conjecture 2 for three agents: every instance with $n = 3$ and negative dichotomous valuations admits an envy-free solution with a subsidy of 0 or 1 per agent, computable in polynomial time (Theorem 11). The proof completes an envy-free *partial* allocation, taken from Tao et al. [TWYZ23], by placing the residual chores and then reassigning which agent receives which bundle; the subsidy bound follows from a bound on the total excess of that reassignment. For general n the conjecture remains open. Section 2 fixes the model and records what transfers from the goods theory unchanged; Section 3 contains the proof.

Related Work. The subsidy line for goods is described above. Within it, the dichotomous subclass has attracted separate attention: Halpern and Shah [HS19] settle binary additive valuations with a total subsidy of $n - 1$, Goko et al. [GIK+21] obtain the analogous bound for binary submodular valuations together with a strategy-proof mechanism, and Barman et al. [BKNS22] subsume both by dropping every structural assumption beyond binary marginals. Dichotomous preferences are themselves a well-studied restriction in social choice [BMS05, BEF21]. On the computational side, Caragiannis and Ioannidis [CI21] show that minimising the subsidy is NP-hard and give a fully polynomial-time approximation scheme for a constant number of agents, and Narayan et al. [NSV21] study the welfare cost of achieving envy-freeness with transfers.

The paper closest to the present setting is that of Kawase et al. [KMS+24], the only work in the envy-freeness-with-subsidy line that admits chores at all. Working with doubly monotone valuations under the two-sided normalisation, it decouples the two jobs that [BDN+20] performed together: rather than constructing a specific allocation and showing that it is cheap to subsidise, it takes *any* EF1 allocation as input and bounds the subsidy needed to convert it, obtaining $n - 1$ per agent and $n(n - 1)/2$ in total. Since EF1 allocations are known to exist for doubly monotone valuations, the wider class comes along at no cost. That reduction is what makes the present question well-posed rather than open-ended, and it is the baseline we are trying to beat.

The complementary repair — no money, weaker fairness notion — is pursued for binary valuations by Bu et al. [BSY23], who show that EFX allocations always exist and can be computed in polynomial time for general binary valuations, thus extending the matroid-rank result of Babaioff et al. [BEF21]. Under binary valuations one therefore has a clean dichotomy on the goods side: either pay 0 or 1 per agent and obtain exact envy-freeness [BKNS22], or pay nothing and obtain EFX [BSY23]. Whether either branch survives the passage to chores is open; this paper addresses the first.

The chores side. Work on indivisible chores has largely developed alongside rather than inside the subsidy line, and the two have met less often than one might expect. Where money has been brought to chores, it has been in the service of *proportionality* rather than envy-freeness: the question studied is the total subsidy needed to bring every agent down to her proportional share of the burden, for which the best bounds under additive disutilities normalised to at most 1 per chore are $n/4$ and then $n/3 - 1/6$, recently obtained together with Pareto optimality by Garg et al. [GSW25]. That is a different fairness notion and a different accounting; it does not bound the subsidy needed for envy-freeness.

The no-money branch, by contrast, has a direct chore analogue of [BSY23]. Tao et al. [TWYZ23] study chores with binary marginals and show that for binary *additive* costs an allocation that is simultaneously EFX and Pareto optimal always exists and is computable in polynomial time. So on the chores side the second half of the binary dichotomy — pay nothing, obtain EFX — is established at least for additive costs, while the first half, pay 0 or 1 per agent and obtain exact envy-freeness, is what Conjecture 2 asks for. That money is genuinely needed here is not obvious a priori but is known: Bhaskar et al. [BSV21] show that deciding whether an exactly envy-free allocation of chores exists is NP-complete already for binary additive costs, and give a polynomial-time EF1 algorithm for chores and for doubly monotone instances.

Where the two lines have met is on the *additive* side, and recently. Lu et al. [LMS26] prove the goods-and-chores analogue of [BDN+20]: for additive utilities in which each item is a good for some agents and a chore for others, with every $|v_i(g)| \leq 1$, a subsidy of one dollar per agent suffices, the bound is tight, and the allocation is computable in polynomial time — hence $n-1$ in total. This settles the additive chore problem completely.

It does not settle ours, and the reason is exactly the reason [BKNS22] was not a corollary of [BDN+20]. Dichotomous valuations are not additive and additive valuations are not dichotomous; the two classes are incomparable. The four corners are

	goods	chores
additive	[BDN+20]	[LMS26]
dichotomous	[BKNS22]	**Conjecture 2**

each of the three settled corners giving one dollar per agent and $n-1$ in total. The present question is the fourth. Note also that [LMS26] does not displace [KMS+24] as the baseline of Section 1: for valuations that are dichotomous but not additive, $n-1$ per agent from [KMS+24] remains the best published bound.

We are not aware of any treatment of envy-freeness with subsidy for chores under dichotomous costs. This is the outcome of a literature search rather than a proof of novelty — [LMS26] appeared in July 2026, which is a reminder of how quickly the surrounding corners are being filled in — and it should be repeated before the paper is circulated.

2 Notation and Preliminaries

We study the allocation of m indivisible items among n agents, with subsidies. Write $N = [n]$ for the set of agents and M for the set of items, $|M| = m$. Each agent $i \in N$ has a valuation $v_i : 2^M \to \mathbb{R}$ with $v_i(\emptyset) = 0$, and we represent an instance by the tuple $\langle N, M, \{v_i\}_{i \in N} \rangle$. Algorithms operate in the standard *value-oracle* model: for each v_i we assume only an oracle that returns $v_i(S)$ for a queried set $S \subseteq M$, and running time is measured in n, m and the number of oracle calls.

An *allocation* $\mathcal{A} = (A_1, \ldots, A_n)$ is an ordered tuple of pairwise disjoint *bundles*, with A_i assigned to agent i. The allocation is *complete* if $\bigcup_{i \in N} A_i = M$ and *partial* otherwise. The ordering matters throughout: much of the theory below concerns permuting a fixed family of bundles among the agents, and for a permutation $\sigma \in \mathbb{S}_n$ we write $\mathcal{A}_\sigma := (A_{\sigma(1)}, \ldots, A_{\sigma(n)})$

for the allocation in which agent i receives $A_{\sigma(i)}$. The utilitarian welfare of $\mathcal{A}$ is $\mathrm{SW}(\mathcal{A}) = \sum_{i \in N} v_i(A_i)$.

Definition 3 (Envy-Freeness). *An allocation $\mathcal{A} = (A_1, \ldots, A_n)$ is* envy-free *(EF) iff $v_i(A_i) \geq v_i(A_j)$ for all agents $i, j \in N$.*

Envy-free allocations of indivisible items need not exist, so we allow a third party to supplement the allocation with money. Agents have quasi-linear utilities: agent i receiving bundle A_i and subsidy p_i enjoys utility $v_i(A_i) + p_i$, with money entering linearly and at the same exchange rate for every agent.

Definition 4 (Envy-Free Solution). *An allocation $\mathcal{A} = (A_1, \ldots, A_n)$ and a subsidy vector $p = (p_1, \ldots, p_n) \in \mathbb{R}^n_+$ constitute an* envy-free solution *$(\mathcal{A}, p)$ iff $v_i(A_i) + p_i \geq v_i(A_j) + p_j$ for all $i, j \in N$.*

An allocation $\mathcal{A}$ is *envy-freeable* if some $p \in \mathbb{R}^n_+$ makes $(\mathcal{A}, p)$ an envy-free solution; this is a property of the allocation alone. Following [BKNS22] we write $\mathrm{M}(p) := \{i \in N \mid p_i \geq p_j \text{ for all } j \in N\}$ for the set of maximally subsidised agents.

We will occasionally need the standard relaxation of envy-freeness. Note that two inequivalent forms of it appear in this literature: the goods-only form of [HS19], which removes one item from the envied bundle, and the two-sided form used by [KMS+24], which removes one item from *either* bundle. They coincide when all items are goods and differ once chores are present, and only the two-sided form supports the doubly monotone results. We adopt the two-sided form.

2.1 Negative Dichotomous Valuations

Definition 5 (Negative Dichotomous Valuation). *A valuation $v_i : 2^M \to \mathbb{R}$ with $v_i(\emptyset) = 0$ is* negative dichotomous *iff*

$$v_i(S) - v_i(S \cup \{g\}) \in \{0, 1\} \qquad \textit{for every } S \subseteq M \textit{ and every } g \in M \setminus S,$$

equivalently iff every marginal $\Delta_i^+(S, g) := v_i(S \cup \{g\}) - v_i(S)$ lies in $\{-1, 0\}$.

Every item is thus a chore for every agent: v_i is non-increasing, so $v_i(S) \geq v_i(T)$ whenever $S \subseteq T$. We stress that nothing further is assumed — the valuations need not be additive, submodular, or subadditive — which is the generality at which [BKNS22] works on the goods side.

It is often more convenient to work with the non-negative *cost form*

$$c_i := -v_i, \qquad \text{so} \qquad c_i(\emptyset) = 0, \quad c_i(S) \leq c_i(T) \text{ for } S \subseteq T, \quad c_i(S \cup \{g\}) - c_i(S) \in \{0, 1\}.$$

Definition 6 (Dichotomous cost function). *A set function $c : 2^M \to \mathbb{R}$ is* dichotomous *if $c(\emptyset) = 0$ and every marginal $c(S \cup \{g\}) - c(S)$ lies in $\{0, 1\}$.*

Monotonicity is not a separate hypothesis: marginals in $\{0, 1\}$ are non-negative, so a dichotomous c is automatically non-decreasing. We nonetheless list it above because it is the property most often used directly.

The correspondence $v_i \leftrightarrow c_i$ is a bijection between negative dichotomous valuations and dichotomous cost functions in the sense of [BKNS22]: the class we study is exactly the pointwise negation of the class they study. In cost form an envy-free solution is the requirement $c_i(A_i) - p_i \le c_i(A_j) - p_j$ for all i, j, read as "my own burden net of my compensation is no worse than what I perceive of yours". We use whichever of v_i and c_i is clearer locally and always state which.

2.2 The Envy Graph and the Halpern–Shah Characterisation

For an allocation $\mathcal{A}$, the *envy graph* $G_{\mathcal{A}}$ is the complete weighted directed graph on vertex set N in which the arc (i, j) carries weight

$$w_{\mathcal{A}}(i, j) \;:=\; v_i(A_j) - v_i(A_i) \;=\; c_i(A_i) - c_i(A_j),$$

the envy that agent i has towards agent j. The weight of a directed path P is $w_{\mathcal{A}}(P) = \sum_{(i,j)\in P} w_{\mathcal{A}}(i, j)$, and $\ell_{\mathcal{A}}(i)$ denotes the maximum weight of any path in $G_{\mathcal{A}}$ starting at i, the empty path of weight 0 included.

The two results below are the engine of the entire subsidy line. Both are stated by Halpern and Shah [HS19] for additive valuations, but their proofs use only the arc weights and permutations of the bundles — neither additivity nor monotonicity nor any sign condition on v_i enters — and [BKNS22] invokes them in exactly this generality for (non-additive) dichotomous valuations. They therefore apply verbatim to negative dichotomous valuations, and we use them without further comment.

Theorem 7 ([HS19])**.** *For any allocation $\mathcal{A} = (A_1, \dots, A_n)$, the following are equivalent.*

(i) $\mathcal{A}$ is envy-freeable.

(ii) $\mathcal{A}$ maximises utilitarian welfare across all reassignments of its own bundles among the agents: for every permutation σ over N, $\sum_{i\in N} v_i(A_i) \ge \sum_{i\in N} v_i(A_{\sigma(i)})$.

(iii) The envy graph $G_{\mathcal{A}}$ has no positive-weight directed cycle.

Theorem 8 ([HS19])**.** *For any envy-freeable allocation $\mathcal{A}$, the subsidy vector p^* given by $p^*_i := \ell_{\mathcal{A}}(i)$ realises an envy-free solution $(\mathcal{A}, p^*)$, and $p^*_i \le p_i$ for every $i \in N$ and every p such that $(\mathcal{A}, p)$ is an envy-free solution. It can be computed in strongly polynomial time by running an all-pairs longest-path computation on $G_{\mathcal{A}}$.*

Condition (ii) of Theorem 7 has the standing consequence that an envy-freeable allocation can always be manufactured from an arbitrary one: fix the bundles and reassign them according to a maximum-weight perfect matching between agents and bundles. What is at issue is never the existence of an envy-freeable allocation, only the size of the subsidy that Theorem 8 then charges for it.

Specialising to our model gives the following, which we use throughout.

Observation 9 (Integrality)**.** *Let $\{v_i\}_{i\in N}$ be negative dichotomous. Then $v_i(S) \in \mathbb{Z}_{\le 0}$ for every i and every $S \subseteq M$; every arc weight $w_{\mathcal{A}}(i, j)$ is an integer; and for every envy-freeable allocation $\mathcal{A}$ the minimum subsidy vector satisfies $p^* \in \mathbb{Z}^n_{\ge 0}$.*

Proof Fix i and $S = \{g_1, \ldots, g_k\}$. Telescoping from $v_i(\emptyset) = 0$ along any ordering of S writes $v_i(S)$ as a sum of k marginals, each of which lies in $\{-1, 0\}$ by Definition 5; hence $v_i(S) \in \mathbb{Z}$ and $-k \le v_i(S) \le 0$. Consequently each arc weight $w_{\mathcal{A}}(i, j) = v_i(A_j) - v_i(A_i)$ is a difference of integers, and the weight of any directed path, being a finite sum of arc weights, is an integer. By Theorem 8, $p_i^* = \ell_{\mathcal{A}}(i)$ is the maximum such weight over paths starting at i, and it is non-negative because the empty path has weight 0. □

Observation 9 is what makes the target statement — a subsidy vector in $\{0,1\}^n$ — a statement about a *bound* rather than about rounding: once p^* is known to be integral, showing $p_i^* \le 1$ for every i is the same as showing $p^* \in \{0,1\}^n$. The same observation underlies the corresponding step in [BKNS22], where the arc weights of a dichotomous instance are likewise integers.

A second normalisation costs nothing and removes a clause from the target.

Proposition 10 (Some agent is always unpaid)**.** *For every envy-freeable allocation $\mathcal{A}$ there is an agent h with $p_h^* = 0$. Consequently, if $p^* \in \{0,1\}^n$ then $\sum_{i \in N} p_i^* \le n - 1$.*

Proof By Theorem 7(iii) the envy graph $G_{\mathcal{A}}$ has no positive-weight cycle, so a walk of maximum weight may be taken simple and $\ell_{\mathcal{A}}$ is finite; it is non-negative because the empty path qualifies. Choose a path P attaining $\max_i \ell_{\mathcal{A}}(i)$ and let h be its final vertex. If $\ell_{\mathcal{A}}(h) > 0$ there is a positive-weight walk Q leaving h, and P followed by Q is a walk from P's start of weight $w_{\mathcal{A}}(P) + w_{\mathcal{A}}(Q) > w_{\mathcal{A}}(P)$, contradicting the maximality of P among walks. Hence $\ell_{\mathcal{A}}(h) = 0$. The second claim is immediate: at most $n - 1$ coordinates of p^* can equal 1. □

Proposition 10 justifies the word "hence" in Conjecture 2: the bound on the *total* is not an independent requirement but a consequence of the per-agent one. The conjecture is therefore a single-clause statement,

$$\exists \mathcal{A} \; : \quad \max_{i \in N} \ell_{\mathcal{A}}(i) \;\le\; 1,$$

and it is this form that the rest of the report works with.

3 The Conjecture for Three Agents

We prove Conjecture 2 for $n = 3$. No assumption is made on the cost functions beyond Definition 6, and none on the number of chores.

Theorem 11. *For every instance with three agents and negative dichotomous valuations there is an envy-free solution $(\mathcal{A}, p)$ with $p \in \{0,1\}^3$, hence with total subsidy at most 2, computable in polynomial time given value-oracle access.*

We work in cost form throughout and write $c_i(g \mid S) := c_i(S \cup \{g\}) - c_i(S) \in \{0,1\}$ for a marginal. The proof completes an envy-free *partial* allocation supplied by the following external result; recall from Section 2 that Definition 3 and Definition 4 apply verbatim to partial allocations.

Theorem 12 ([TWYZ23])**.** *For any number n of agents and any dichotomous cost functions there is a polynomial-time algorithm producing a partial allocation $(A_1, \ldots, A_n)$ that is envy-free with $p = 0$ and leaves at most $n - 1$ chores unassigned.*

At $n = 3$ this yields an envy-free partial allocation together with a residue $R := M \setminus (A_1 \cup A_2 \cup A_3)$ of at most two chores. If $R = \emptyset$ the allocation is already complete and Theorem 11 holds with $p = 0$. The cases $|R| = 1$ and $|R| = 2$ are settled in Section 3.2 and Section 3.3 respectively, using the quantity introduced next.

3.1 Excess

Definition 13 (Excess)**.** *Fix bundles $Y_1, \ldots, Y_n$ and an assignment $\sigma \in \mathbb{S}_n$, agent a receiving $Y_{\sigma(a)}$. Put $\beta_a := \min_j c_a(Y_j)$ and*

$$\varepsilon_a(\sigma) \;:=\; c_a(Y_{\sigma(a)}) - \beta_a \;\geq\; 0.$$

The total excess *of σ is $\sum_{a \in N} \varepsilon_a(\sigma)$.*

Lemma 14. *Let $\mathcal{A}$ be the allocation assigning $Y_{\sigma(a)}$ to a. Every simple directed path in $G_{\mathcal{A}}$ has weight at most $\sum_a \varepsilon_a(\sigma)$; hence $\ell_{\mathcal{A}}(i) \leq \sum_a \varepsilon_a(\sigma)$ for every $i \in N$.*

Proof Each arc leaving a has weight $w_{\mathcal{A}}(a, b) = c_a(Y_{\sigma(a)}) - c_a(Y_{\sigma(b)}) \leq c_a(Y_{\sigma(a)}) - \beta_a = \varepsilon_a(\sigma)$, because β_a is a minimum over all of $Y_1, \ldots, Y_n$. A simple path leaves each agent at most once, and the empty path has weight 0. □

Lemma 15. *For every assignment σ, $\sum_a c_a(Y_{\sigma(a)}) = \sum_a \beta_a + \sum_a \varepsilon_a(\sigma)$. Since $\sum_a \beta_a$ does not depend on σ, an assignment minimises total cost if and only if it minimises total excess.*

Proof Sum Definition 13 over $a \in N$. □

Corollary 16. *Let $Y_1, \ldots, Y_n$ be bundles admitting some assignment of total excess at most 1, and let π minimise $\sum_a c_a(Y_{\pi(a)})$. Then the allocation $\mathcal{A}$ assigning $Y_{\pi(a)}$ to a admits an envy-free solution with $p^* \in \{0, 1\}^n$.*

Proof By Lemma 15, π has total excess at most 1. In cost form $\sum_a v_a(Y_{\pi(a)}) = -\sum_a c_a(Y_{\pi(a)})$, so minimising total cost across reassignments of $Y_1, \ldots, Y_n$ is exactly condition (ii) of Theorem 7; hence $\mathcal{A}$ is envy-freeable. Its minimal subsidy is $p_i^* = \ell_{\mathcal{A}}(i)$ by Theorem 8, which is at most 1 by Lemma 14, and integral and non-negative by Observation 9. □

3.2 One unallocated chore

The following holds for every number of agents; only the case $n = 3$ is needed for Theorem 11.

Theorem 17. *Let $(A_1, \ldots, A_n)$ be an envy-free partial allocation leaving exactly one chore e unassigned. Fix any $m \in N$, put $Y_m := A_m \cup \{e\}$ and $Y_j := A_j$ for $j \neq m$, and let π minimise $\sum_a c_a(Y_{\pi(a)})$. Then the allocation assigning $Y_{\pi(a)}$ to a admits an envy-free solution with $p^* \in \{0, 1\}^n$.*

Proof By Corollary 16 it suffices to exhibit one assignment of total excess at most 1; the identity assignment is one.

Let $a \neq m$. For every j,

$$c_a(Y_a) \;=\; c_a(A_a) \;\leq\; c_a(A_j) \;\leq\; c_a(Y_j),$$

the first inequality by envy-freeness of the partial allocation and the second by monotonicity, since $A_j \subseteq Y_j$. Hence $\beta_a = c_a(Y_a)$ and $\varepsilon_a(\mathrm{id}) = 0$.

For $a = m$ and every j,

$$c_m(Y_m) \;=\; c_m(A_m) + c_m(e \mid A_m) \;\leq\; c_m(A_m) + 1 \;\leq\; c_m(A_j) + 1 \;\leq\; c_m(Y_j) + 1,$$

using Definition 6, then envy-freeness, then monotonicity. Taking the minimum over j gives $c_m(Y_m) \leq \beta_m + 1$, that is $\varepsilon_m(\mathrm{id}) \leq 1$. □

3.3 Two unallocated chores

This case uses more than the statement of Theorem 12: it uses the state in which its algorithm halts. That algorithm maintains an envy-free partial allocation $(A_1, \dots, A_n)$ together with the *equality graph*

$$G = (N, E), \qquad (i, j) \in E \iff i \neq j \;\text{ and }\; c_i(A_i) = c_i(A_j), \tag{1}$$

and repeats three rules in the order below, halting only when none applies.

(R1) If some unassigned chore e and some agent i satisfy $c_i(e \mid A_i) = 0$, assign e to i.

(R2) If some unassigned chore e and some arc $(i, j) \in E$ lying on a directed cycle of G satisfy $c_i(e \mid A_j) = 0$, rotate the bundles along that cycle and assign e to i.

(R3) Otherwise select a strongly connected component S of G with no arc of E leaving S. If at least $|S|$ chores are unassigned, assign one to each agent of S; otherwise halt.

Lemma 18. *Suppose the algorithm of Theorem 12, run on three agents, halts with partial allocation (A_1, A_2, A_3) and residue R of size 2. Then, with G as in (1):*

(a) G is strongly connected;

(b) $c_i(e \mid A_i) = 1$ for every $i \in N$ and every $e \in R$;

(c) $c_i(e \mid A_j) = 1$ for every arc $(i, j) \in E$ and every $e \in R$.

Proof Chores remain unassigned, so the algorithm halted at the final clause of (R3); in particular neither (R1) nor (R2) applied at the halting state.

(b) If $c_i(e \mid A_i) = 0$ for some i and some $e \in R$, rule (R1) would apply. Marginals lie in $\{0, 1\}$, hence the value is 1.

(a) The component S selected by (R3) satisfies $|R| < |S|$. As $|R| = 2$ this forces $|S| = 3$, so $S = N$; and S is strongly connected by construction.

(c) Let $(i, j) \in E$. By (a) there is a directed path in G from j to i; choosing one with no repeated vertex and appending the arc (i, j) produces a directed cycle of G through (i, j). If $c_i(e \mid A_j) = 0$ for some $e \in R$, rule (R2) would apply to that arc and cycle. □

Lemma 19. *In the situation of Lemma 18, for every $a, j \in N$ and every $e \in R$,*

$$c_a(A_j \cup \{e\}) \;\geq\; c_a(A_a) + 1.$$

Proof Write $c_a(A_j \cup \{e\}) = c_a(A_j) + c_a(e \mid A_j)$. Envy-freeness of the partial allocation gives $c_a(A_j) \geq c_a(A_a)$, and both sides are integers by Observation 9. If that inequality is strict the claim follows, marginals being non-negative. If it is an equality then either $j = a$, and $c_a(e \mid A_a) = 1$ by Lemma 18(b); or $j \neq a$, in which case $(a, j) \in E$ by (1) and $c_a(e \mid A_j) = 1$ by Lemma 18(c). □

Theorem 20. *Let (A_1, A_2, A_3) be the partial allocation at which the algorithm of Theorem 12 halts with residue $R = \{e_1, e_2\}$. Assign e_1 and e_2 to two* distinct *bundles, and assign the three resulting bundles to the three agents by a minimum-cost matching. The resulting allocation admits an envy-free solution with $p^* \in \{0, 1\}^3$.*

Proof Write Y_1, Y_2, Y_3 for the bundles after placement: two are *augmented*, carrying one chore of R each, and one is *clean*. Put $L := \sum_a c_a(A_a)$ and, for an assignment σ,

$$g_a(\sigma) \;:=\; c_a(Y_{\sigma(a)}) - c_a(A_a) \;\geq\; 0,$$

non-negative because $c_a(Y_{\sigma(a)}) \geq c_a(A_{\sigma(a)}) \geq c_a(A_a)$ by monotonicity and envy-freeness of the partial allocation.

The minimum of $\sum_a c_a(Y_{\sigma(a)})$ is exactly $L + 2$. For the lower bound, fix σ: the two agents receiving augmented bundles have $g_a(\sigma) \geq 1$ by Lemma 19, and the third has $g_a(\sigma) \geq 0$. For the upper bound, hand each bundle to its original owner: the two agents whose own bundles were augmented pay $c_a(A_a) + 1$ by Lemma 18(b), and the third pays $c_a(A_a)$.

Hence at a minimum-cost assignment π the excesses are pinned: $\sum_a g_a(\pi) = 2$ with $g_a(\pi) \geq 1$ on each of the two augmented-bundle holders, so both equal 1 exactly and the agent c holding the clean bundle has $g_c(\pi) = 0$.

Let a and b be the augmented-bundle holders, and recall $w(x, y) = c_x(Y_{\pi(x)}) - c_x(Y_{\pi(y)})$.

- $w(a, b) = \big(c_a(A_a) + 1\big) - c_a(Y_{\pi(b)}) \leq 0$: the first term because $g_a(\pi) = 1$, and $c_a(Y_{\pi(b)}) \geq c_a(A_a) + 1$ by Lemma 19, the bundle $Y_{\pi(b)}$ being augmented. Symmetrically $w(b, a) \leq 0$.
- $w(a, c) = \big(c_a(A_a) + 1\big) - c_a(Y_{\pi(c)}) \leq 1$, since $Y_{\pi(c)}$ is clean and therefore costs a at least $c_a(A_a)$. Symmetrically $w(b, c) \leq 1$.
- $w(c, x) = c_c(A_c) - c_c(Y_{\pi(x)}) \leq 0$ for both x, since $g_c(\pi) = 0$ gives the first term and every bundle costs c at least $c_c(A_c)$.

Every arc therefore has weight at most 0 except the two arcs entering c, which have weight at most 1. A simple path enters c at most once, every arc leaving c has weight at most 0, and every other arc available to it has weight at most 0; hence no simple path has weight above 1. The assignment is minimum-cost, so the allocation is envy-freeable by Theorem 7, and its minimal subsidy is $p_i^* = \ell(i) \leq 1$ by Theorem 8, integral and non-negative by Observation 9. □

Remark 21. Theorem 20 appeals to the halting state of the algorithm of Theorem 12, not merely to that theorem's statement, and the distinction is not cosmetic. The hypotheses of Lemma 18 do not follow from "envy-free partial allocation with two unassigned chores, each of unit marginal on its holder's own bundle". Strong connectivity of G is available only because two is the *maximum* number of chores the algorithm may leave at $n = 3$, which is what forces the component selected by (R3) to be the whole agent set.

3.4 Proof of the main theorem

Proof of Theorem 11 Run the algorithm of Theorem 12 on the three cost functions c_1, c_2, c_3, obtaining an envy-free partial allocation (A_1, A_2, A_3) with residue R of size at most 2.

If $|R| = 0$ the allocation is complete and, being envy-free, $(\mathcal{A}, \mathbf{0})$ is an envy-free solution. If $|R| = 1$, apply Theorem 17. If $|R| = 2$, apply Theorem 20. In each case the resulting complete allocation admits an envy-free solution with $p^* \in \{0, 1\}^3$, and Proposition 10 bounds the total by 2.

For the running time, the algorithm of Theorem 12 is polynomial and returns (A_1, A_2, A_3) and R. The completion places at most two chores into bundles and then computes a minimum-cost matching between three agents and three bundles, which the Hungarian method performs in constant time after the nine oracle calls determining $c_a(Y_j)$. Hence the whole procedure is polynomial. □

References

[BCE+16] Felix Brandt, Vincent Conitzer, Ulle Endriss, Jérôme Lang, and Ariel D. Procaccia. *Handbook of Computational Social Choice*. Cambridge University Press, 1st edition, 2016.

[BDN+20] Johannes Brustle, Jack Dippel, Vishnu V. Narayan, Mashbat Suzuki, and Adrian Vetta. One dollar each eliminates envy. In *Proceedings of the 21st ACM Conference on Economics and Computation (EC)*, pages 23–39, 2020.

[BEF21] Moshe Babaioff, Tomer Ezra, and Uriel Feige. Fair and truthful mechanisms for dichotomous valuations. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 35. AAAI, 2021.

[BKNS22] Siddharth Barman, Anand Krishna, Y. Narahari, and Soumyarup Sadhukhan. Achieving envy-freeness with limited subsidies under dichotomous valuations, 2022.

[BMS05] Anna Bogomolnaia, Hervé Moulin, and Richard Stong. Collective choice under dichotomous preferences. *Journal of Economic Theory*, 122(2):165–184, 2005.

[BSV21] Umang Bhaskar, A. R. Sricharan, and Rohit Vaish. On approximate envy-freeness for indivisible chores and mixed resources. In *Approximation, Randomization, and Combinatorial Optimization (APPROX/RANDOM)*, LIPIcs, 2021.

[BSY23] Xiaolin Bu, Jiaxin Song, and Ziqi Yu. EFX allocations exist for binary valuations. In *Proceedings of the International Conference on Frontiers of Algorithmic Wisdom (FAW)*, 2023.

[Bud11] Eric Budish. The combinatorial assignment problem: Approximate competitive equilibrium from equal incomes. *Journal of Political Economy*, 119(6):1061–1103, 2011.

[CI21] Ioannis Caragiannis and Stavros Ioannidis. Computing envy-freeable allocations with limited subsidies. In *Workshop on Internet and Network Economics (WINE)*, 2021.

[End17] Ulle Endriss. *Trends in Computational Social Choice*. Lulu.com, 2017.

[Fol67] Duncan Karl Foley. Resource allocation and the public sector. *Yale Economic Essays*, 7(1):45–98, 1967.

[GIK+21] Hiromichi Goko, Ayumi Igarashi, Yasushi Kawase, Kazuhisa Makino, Hanna Sumita, Akihisa Tamura, Yu Yokoi, and Makoto Yokoo. Fair and truthful mechanism with limited subsidy, 2021.

[GSW25] Jugal Garg, Eklavya Sharma, and Xiaowei Wu. Proportional and Pareto-optimal allocation of chores with subsidy, 2025.

[HS19] Daniel Halpern and Nisarg Shah. Fair division with subsidy. In *Proceedings of the 12th International Symposium on Algorithmic Game Theory (SAGT)*, volume 11801 of *Lecture Notes in Computer Science*, pages 374–389. Springer, 2019.

[KMS+24] Yasushi Kawase, Kazuhisa Makino, Hanna Sumita, Akihisa Tamura, and Makoto Yokoo. Towards optimal subsidy bounds for envy-freeable allocations. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 2024.

[LMMS04] Richard J. Lipton, Evangelos Markakis, Elchanan Mossel, and Amin Saberi. On approximately fair allocations of indivisible goods. In *Proceedings of the 5th ACM Conference on Electronic Commerce (EC)*, pages 125–131, 2004.

[LMS26] Xinhang Lu, Simon Mackenzie, and Mashbat Suzuki. Optimal subsidy bounds for goods and chores: One dollar each suffices, 2026.

[Mas87] Eric S. Maskin. On the fair allocation of indivisible goods. In G. Feiwel, editor, *Arrow and the Foundations of the Theory of Economic Policy*, pages 341–349. MacMillan, 1987.

[NSV21] Vishnu V. Narayan, Mashbat Suzuki, and Adrian Vetta. Two birds with one stone: Fairness and welfare via transfers. In *International Symposium on Algorithmic Game Theory (SAGT)*, pages 376–390. Springer, 2021.

[TWYZ23] Biaoshuai Tao, Xiaowei Wu, Ziqi Yu, and Shengwei Zhou. On the existence of EFX (and pareto-optimal) allocations for binary chores, 2023. Journal version in Theoretical Computer Science, 2025.

[Var74] Hal Varian. Equity, envy, and efficiency. *Journal of Economic Theory*, 9(1):63–91, 1974.